

A SOUTH AFRICAN BRAND

ORIGIN

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Checklists, Tickets & the SMF

— *how they wire together.*

Every paper SOP that recurs becomes a **Checklist Template**. The Schedule Engine fires it on cadence, the Ticket System carries it to the right person, the Sign-off Chain stamps it, and the completed instance becomes **SMF evidence**. One substrate. Three primitives. SAHPRA-replayable.

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SOPS NEED CHECKLISTS

3

PRIMITIVES

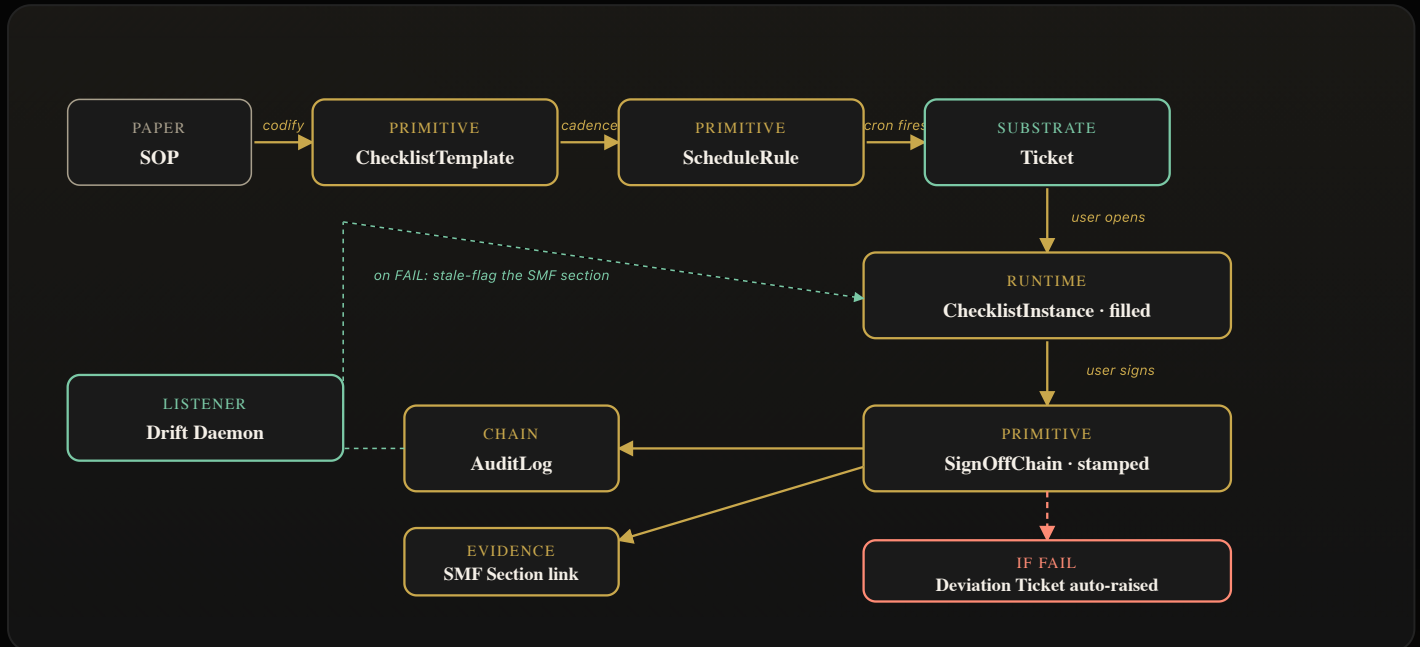
1

AUDIT CHAIN

01 The model in one breath

SINGLE SUBSTRATE · SEVEN NODES · TWO FAILURE PATHS

Every paper SOP that recurs becomes a **Checklist Template**. The **Schedule Engine** fires the template on cadence (daily / weekly / monthly / per-batch), spawning a **Ticket** whose body IS the checklist. The user taps through the checks, the **Sign-off Chain** stamps it, and the completed instance is filed as **SMF evidence** linked to the section(s) it informs. The Drift Daemon listens for any check that fails and either auto-raises a **Deviation ticket** or marks an **SMF section stale**.



02 Where checklists are needed

CADENCE-DRIVEN · 31 OF 76 SOPS

Checklists are the right UX whenever an SOP says "do these *N* things in order" or "verify these *N* points are true." From the 76-SOP catalogue, **31 need checklists**. They group naturally by cadence — and that grouping is also how the Schedule Engine fires them.

CADENCE Daily	4-FPH-009 hygiene & PPE · 6-SEC-001 perimeter walk · 4-FPH-025 S6 in / out (transactional)	Floor staff sign in by tapping through the steps
CADENCE Weekly	4-FPH-008 housekeeping · 6-SEC-002 CCTV / alarm test · 4-SEC-010 genset test · 3-INV-008 Logtag download	Friday cockpit – one tap to run all four
CADENCE Monthly	4-FPH-010 pest control · 4-FPH-021 expiry sweep · 4-FPH-026 spillage kit · 4-FPH-027 month-end · 4-FPH-018 stability pulls	Monthly cockpit reminds RP + FM
CADENCE Quarterly	2-LEG-003 complaints review · 6-SEC-004 IT access review	Cross-functional review forms
CADENCE Annual	2-LEG-006 SIA (9-section heavyweight) · 2-LEG-005 vendor audit · 2-LEG-007 courier audit · 4-FPH-004 risk register · 4-SEC-009 fire safety · 2-LEG-001 director ethics declaration	Year-plan widget; SIA is its own wizard
CADENCE Per-batch / Event	All 3-CUL-001..010 stage records · 4-FPH-002 deviation · 4-FPH-015 finished-product release · 4-FPH-024 stock receiving	Triggered by the batch reaching the stage, not by the clock
CADENCE Per-licence	1-DOC-001 3-yr SOP review · 1-DOC-003 5-yr archive purge · 2-LEG-001 90-day licence renewal	Long-period reminders surface on the AR's queue

Where you don't need a checklist

SOPs that are *reference policies* — 1-DOC-004 organogram, 2-LEG-013 conditions of employment, 2-LEG-016 quality manual. These are documents that get read and signed once per revision, not workflows you tap through.

03 How a checklist links to a ticket

LIFECYCLE OF A DAILY PERIMETER WALK • 6-SEC-001

The ticket system is already the substrate. A checklist is just a **structured ticket body** with one row per check. Here is what one day in the life of a perimeter walk looks like, end-to-end.

- 1 **02:00 • CRON FIRES**
`ScheduleRule` for "Perimeter walk" creates a `ChecklistInstance` with `dueAt = 06:00`, assignee = on-shift Security.
- 2 **02:00 • TICKET SPAWNED**
A `Ticket` of type `CHECKLIST_DUE` is auto-created, linked to the instance, assigned to Sipho. It appears in his action tray as "Log perimeter walk".
- 3 **05:30 • USER OPENS**
Sipho taps the action button. The `<ChecklistRunner>` opens with 8 items: gate 1, gate 2, fence sections 1-3, motion sensors, CCTV check, fire-extinguisher visibility.
- 4 **05:30 – 05:48 • TAP-THROUGH**
Each tap appends to `ChecklistInstance.results`. If any item is **FAIL** → a **Deviation ticket** auto-spawns linked back to this instance. Photo evidence attaches to the failed step automatically.
- 5 **05:48 • SIGN**
Sipho hits **Sign**. `SignOffChain` stamps `EXECUTOR`. Some checklists need a second signer (RP for S6 register, AR for month-end); this one doesn't.
- 6 **05:48 • CLOSED & CHAINED**
Instance closed. Hash chained into `AuditLog`. Original `CHECKLIST_DUE` ticket auto-resolves. SMF Section **E.1 Site Security** evidence count ticks up by 1.
- 7 **05:55 • SLA ESCALATION (DIDN'T HAPPEN HERE)**
If `completedAt` were still null at SLA T-5min, the ticket would have escalated to FM automatically. Same primitive, opposite outcome.

Same pattern for everything else

Monthly month-end (**4-FPH-027**), annual SIA (**2-LEG-006**), per-batch finished-product release (**4-FPH-015**) — all the same lifecycle. *Only the cadence and the signers differ.*

04 Data shapes · the engineering room view

THREE PRISMA MODELS · ONE REACT WIDGET

The whole system is three primitives in the schema and one runtime component. Below is the minimal viable shape.

```
model ChecklistTemplate {
  id          String   @id @default(cuid())
  sopId       String   // e.g. "6-SEC-001"
  title       String
  version     Int       @default(1)
  items       Json      // [{ id, text, type, requiredEvidence?, failAction }]
  signers     String[]  // ["EXECUTOR", "QA", "RP"]
  smfSectionRefs String[] // SMF sections this informs
}

model ScheduleRule {
  id          String   @id @default(cuid())
  templateId  String
  cadence     Cadence  // DAILY | WEEKLY | MONTHLY | QUARTERLY | ANNUAL
  dayOfWeek   Int?
  dayOfMonth  Int?
  time        String?  // "06:00"
  slaHours    Int
  assigneeRole String   // "SECURITY_OFFICER"
  escalateTo  String?   // "FACILITY_MANAGER"
}

model ChecklistInstance {
  id          String   @id @default(cuid())
  templateId  String
  ticketId    String
  dueAt       DateTime
  completedAt DateTime?
  results     Json      // [{ itemId, value, evidenceUrl?, atTimestamp }]
  signatures  Json      // [{ role, userId, signedAt, hash }]
  smfSectionLinks String[]
}
```

The single React component

`<ChecklistRunner templateId={...} />` handles every checklist family — generic, renders any template, dispatches to the right item-type sub-renderer (`CHECK` · `NUMERIC` · `PHOTO` · `SIG`), posts results, fires the SignOffChain at the end. **Five item types** cover every SOP we read.

05 How checklists map to the SMF

THE CLOSED REGULATORY LOOP · EVIDENCE + PROVENANCE

This is the regulator-grade part. Two relationships matter.

A • Which SMF section does each checklist family inform?

Every checklist template carries `smfSectionRefs: string[]`: the SMF sections whose accuracy depends on this checklist being completed. When a stale flag appears on an SMF section, the SMF Composer reads recent completed checklists for that section as fresh evidence — no manual gathering needed.

CHECKLIST FAMILY	MAPS TO SMF SECTION(S)
Cultivation stage logs (3-CUL-001..010)	C.3.1 Cultivation Areas · C.3.2 Plant Lifecycle
Logtag T & H (3-INV-008 / 4-FPH-019)	C.3.5 Environmental Controls · C.5.2 Calibration
Hygiene & PPE (4-FPH-009)	C.4 Personnel Hygiene
Housekeeping (4-FPH-008)	C.4.3 Cleaning & Sanitation
Pest control (4-FPH-010)	C.4.4 Pest Control
S6 register (4-FPH-025)	C.6 Controlled Substance Handling
Perimeter / vault (6-SEC-001 / 005)	E.1 Site Security
IT access review (6-SEC-004)	E.2 Information System Security
Equipment calibration (4-SEC-009 / 4-SEC-010)	C.1.6 Equipment Register
Annual SIA (2-LEG-006)	A.1 Quality Management — informs the entire SMF

B • Completed checklists feed the sign-off chain

Every `ChecklistInstance` is itself signed by the same `SignOffChain` widget we already built for SMF sections. When the SMF section it informs comes up for re-sign or AR freeze, the regulator can drill: *"show me the last 30 days of perimeter checklists supporting E.1"* → Inspector dashboard returns the list with hashes.

This closes the loop. The SMF stops being a static binder and becomes a live document whose signed sections are continuously backed by signed checklists. SAHPRA doesn't ask *"do you do perimeter walks?"* — they ask *"show me yesterday's, signed, time-stamped, with the failure that triggered ticket TKT-0481."* That answer is one tap from any role.

06 What "functional" means · the smallest meaningful thing

PHASE 0 + 1 · TWO SPRINTS · CUTTABLE TODAY

To move from "we have a plan" to "it actually runs," cut these tickets in this order. Two sprints to first checklists running on real data with real sign-off and real SMF linkage.

01	Prisma migration — ChecklistTemplate , ScheduleRule , ChecklistInstance , ChecklistResult , ChecklistSignature . Append-only on results + signatures.	backend / prisma
02	Schedule Engine daemon — cron (BullMQ or pg-cron) reads ScheduleRule rows and writes ChecklistInstance + Ticket rows. One service file, ~150 lines.	backend / services
03	<ChecklistRunner> component — generic, renders any template, handles CHECK / NUMERIC / PHOTO / SIG item types, posts results, fires SignOff at the end.	frontend / components
04	Seed first 5 checklists — hygiene · perimeter · S6 in / out · housekeeping · monthly expiry sweep. These exercise every item type and signer permutation.	backend / prisma / seed
05	Wire into Maestro — when a user opens chat, the Maestro action tray surfaces pending CHECKLIST_DUE tickets as tap-buttons. Tapping opens the Runner in a modal.	backend / chat
06	Wire into SMF drift — register the 10 mappings on page 06 so a failed pest-control checklist auto-stales C.4.4. Drift Daemon already has the listener; just register the rules.	backend / drift

That's it. Two sprints to working checklists.

Phase 0 + Phase 1 of the SOP Action Plan, scoped narrowly to "checklists with schedule + sign-off + SMF link." After that, the heavyweight wizards (SIA, Recall, TQCRF) are just bigger templates running on the same primitives. Tell me to cut tickets and I'll start in `tnt-za/backend/prisma/schema.prisma` + `tnt-za/backend/src/services/schedule.service.ts` next.